

「請各出題委員繕打考題時，一定要使用此格式，請勿自行變更該格式」謝謝！

「以下為範例出題格式，請各委員出題時，請以下列格式繕打，切勿變更其格式，謝謝！」

1. () 下列何者不屬於 larynx 的解剖構造？

- A) arytenoid
- B) ventricle
- C) cricoid cartilage
- D) epiglottis

(1 答案): C (頭頸癌-Laryngeal cancer)

(1 難易度): 簡單

(1 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 48, p1043; 2019.

(1 註解): P1043, The larynx is divided into the supraglottis, glottis, and subglottis. The supraglottis consists of the epiglottis, false vocal cords, ventricles, aryepiglottic folds, and the arytenoids.

2. () 有關 altered fractionation radiotherapy 在頭頸癌的應用，EORTC 22791, hyperfractionated RT 在口咽癌運用之描述，下列何者為誤？

- A) Altered fractionation RT 在頭頸癌的應用，研究最多的族群是接受 definitive oropharyngeal cancer 的病人。
- B) EORTC 22791 主要收入 T2-3, N0-1 (<3 cm)，非 base of tongue 的口咽癌病人，發現 hyperfractionated RT 比 conventional fraction RT 有較好的局部控制率，但 long-term toxicity 發生率也較高。
- C) Accelerated RT 66 to 70 Gy delivered in 2 Gy daily fractions 6 day a week 的方式相較於 conventional fractionation 也有較好的局部控制，但是 1.8 Gy twice daily to 59.4 Gy 卻沒有。
- D) Simultaneous integrated boost radiotherapy (2.2, 2, and 1.8 Gy to gross tumor and intermediate-risk and low-risk PTVs) 可考慮用於 early-staged oropharyngeal cancer。

(2 答案): B (頭頸癌-口咽癌)

(2 難易度): 中等

(2 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 48, p1013-1014; 2019.

(2 註解): P 1013, Hyperfractionated RT was associated with statistically significant improvements in locoregional control (5-year, 59% vs. 40%) without an increase in long-term toxicity (similar toxicity).

3. () 有關 RT 或 CCRT 後局部復發之口咽癌之描述，下列何者為誤？

- A) 建議手術切除，但可手術切除的病人不多
- B) 手術後有 high-risk features 或無法手術的病人中，20%行再 CCRT 有長期存活率
- C) re-RT 病人，頸動脈破裂機率最高可到 20%

D) SBRT 有較好的腫瘤控制率及較低的長期毒性

(3 答案): **D (頭頸癌-口咽癌)**

(3 難易度): 簡單

(3 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 48, p1023; 2019.

(3 註解): P1023, similar to chemo-reirradiation, high-grade late toxicities including carotid blowout are possible (similar long-term toxicity).

4. () 有關 salivary gland tumor 之描述，下列何者為誤？

- A) Low-grade mucoepidermoid, polymorph low-grade adenocarcinoma, epithelial–myoepithelial, 和 acinic cell carcinomas 皆屬於 low-to-intermediate malignancy。
- B) 有 Her-II overexpression 的 salivary gland tumor 通常預後較好，但若是 high Ki67 或 p53 expression 則預後不好。
- C) PET scan 在 salivary gland 診斷上因為 sensitivity 比較低，故不應該作為常規檢查。
- D) Parotid gland 的神經支配主要由第七對腦神經(CN VII)分支至 superficial and deep lobes

(4 答案): **B (頭頸癌-唾液腺癌)**

(4 難易度): 中等

(4 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 46, p957-962; 2019.

(4 註解): P962, Overexpression of HER2/neu is related with aggressiveness, distant metastases, and poor survival in salivary gland cancer

5. () 有關 adenoid cystic carcinoma 之描述，下列何者為誤？

- A) Adenoid cystic carcinoma 最常見於 submandibular gland，其次是 parotid gland
- B) Perineural invasion 常見於 adenoid cystic carcinoma
- C) Adenoid cystic carcinoma 中，tubular pattern 有較好的預後
- D) Adenoid cystic carcinoma 中，cribriform pattern 預後中等，solid pattern 預後最差

(5 答案): **A (頭頸癌-唾液腺癌)**

(5 難易度): 中等

(5 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 46, p960; 2019.

(5 註解): P960, Adenoid cystic carcinoma is most common in minor salivary glands, followed by the submandibular gland.

6. () 有關下咽癌之描述，下列何者為誤？

- A) 大部分下咽癌病人發現時已是局部晚期，約 25%為臨床三期，50%為臨床四期

- B) 根據 SEER 資料庫，下咽癌 83% 好發於 pyriform sinus，9% 在 posterior pharyngeal wall，4% 在 postcricoid region
- C) 下咽癌病人容易復發在頸部淋巴結 level II to IV 及 retropharyngeal nodes。同側復發率大於對側。放射治療頸部淋巴結建議同側照射即可
- D) 下咽癌遠端轉移率約 16-25%，最常轉移處為肺部

(6 答案): C (頭頸癌-下咽癌)

(6 難易度): 中等

(6 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 49, p1029; 2019.

(6 註解): P1029, In light of cross-draining lymphatics, there is a significant risk of bilateral cervical node metastasis.

7. () 有關 locally advanced 頭頸癌放射治療之描述，下列何者為誤？

- A) 建議同步化學放射治療加上標靶治療
- B) 同步化學放射治療比 conventional RT alone 效果好
- C) 同步化學放射治療比 RT alone with modified fractionation regimen 效果好
- D) 同步化學放射治療有較高的急性及慢性毒性

(7 答案): A (頭頸癌-locally advanced H&N cancer)

(7 難易度): 中等

(7 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 43, p892&894; 2019.

(7 註解): P894, RTOG 0522: No differences were observed in any survival outcome including progression-free survival, overall survival, or patterns of disease failure.

8. () 頸動脈狹窄是頭頸癌放射治療可能致命的副作用，建議高風險病人規律追蹤頸動脈超音波，下列何者不是頸動脈狹窄高風險？

- A) 男性
- B) 年齡大於 60 歲
- C) 抽菸
- D) 高血壓

(8 答案): A (頭頸癌-NPC)

(8 難易度): 簡單

(8 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 44, p929; 2019.

(8 註解): P929, Some have advocated for routine duplex ultrasound screening for high-risk patients (older than 60 years, smoking, hypertension, hypercholesterolemia, cerebrovascular symptoms).

9. () 鼻咽癌放射治療造成的副作用，下列何者發生率最低？

- A) 顳葉壞死 (temporal lobe necrosis)

- B) 聽力缺損 (hearing loss)
- C) 內分泌失調 (endocrine dysfunction)
- D) 次發性腫瘤 (second malignancies)

(9 答案): D (頭頸癌-NPC)

(9 難易度): 簡單

(9 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 44, p928-929; 2019.

(9 註解): P929, Radiation-induced malignancy is rare, with an incidence of 0.04% and latency period >10 years.

10. () 口腔癌最重要的預後因子為下列何者？

- A) 淋巴轉移 (cervical metastases)
- B) 腫瘤厚度 (tumor thickness)
- C) 腫瘤侵犯深度 (depth of invasion)
- D) 神經周邊侵犯 (perineural invasion)

(10 答案): A (頭頸癌-Oral cavity)

(10 難易度): 簡單

(10 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 47, p998; 2019.

(10 註解): P998, The most significant prognostic factor for outcome in oral cavity carcinoma is the presence of cervical metastases.

11. () 有關放射線治療對眼睛構造的影響敘述，下列何者錯誤？

- A) 放射線劑量 $\geq 30\text{Gy}$ 時容易造成 acute conjunctivitis, 治療時若讓病人張開眼睛並選用 megavoltage 能量射線可減少 conjunctivitis 產生的機會。
- B) 造成視網膜 (retina)傷害的 threshold dose 大約是 30-35Gy, 若劑量超過 50Gy 時 radiation retinopathy 的機會急遽增加。
- C) Radiation-induced optic neuropathy 在視神經接受 <50 Gy 時很少發生, 但在 >60Gy 時發生率會顯著上升, 且 fraction size 的影響比 total dose 更為重要。
- D) Lens 對放射線非常敏感, 通常超過 6.5Gy 時日後形成白內障的機會大約有 1/3, 且無特殊治療方式。

(11 答案): D (頭頸癌-tumors of the eye and orbit)

(11 難易度): 中等

(11 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 41, p860-861; 2019.

P860, In adults, higher doses are associated with radiation- induced cataract ... after 6.51 to 11.5Gy, there is a 66% risk with a latent period of 4 years. The corrective treatment for RT-induced cataract is surgery, which yields excellent results.

12. () 有關長在耳朵的惡性腫瘤，下列敘述何者正確？

- A) 在外耳 (external ear)最常見的惡性腫瘤為鱗狀細胞癌 (squamous cell carcinoma)。
- B) 在外耳道(external auditory canal)的惡性腫瘤很少淋巴轉移。
- C) 發生在中耳(middle ear)及顳骨(temporal bone)的惡性腫瘤通常需先手術切除，再輔以術後放射線治療已達局部控制。
- D) 耳蝸(cochlea)的平均劑量(mean dose) <55Gy 方可降低 sensorineural hearing loss。

(12 答案): C (頭頸癌-Ear)

(12 難易度): 難

(12 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 41, p877-882; 2019.

P880, In management of temporal bone tumors originating from the middle ear and mastoid area, surgical options include Postoperative radiation therapy is recommended and increases local tumor control.

13. () 有關長在鼻腔(nasal cavity)及副鼻竇(paranasal sinus)的惡性腫瘤，下列敘述何者正確？

- A) 上頤竇(maxillary sinus)惡性腫瘤若為 suprastructure tumors 則預後較佳，治療效果也較好。
- B) 在鼻腔的惡性腫瘤很少淋巴轉移，通常不需預防性淋巴區域的照射。
- C) 為了器官保留的目的，副鼻竇腫瘤通常以 definitive radiotherapy +/- concurrent chemotherapy 為治療首選。
- D) 為了達到較好的局部控制，hyperfractionated radiotherapy 也被運用在鼻腔及副鼻竇的放射線治療。

(13 答案): B (頭頸癌-cancer of the nasal cavity and paranasal sinuses)

(13 難易度): 中等

(13 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 41, p936-947; 2019.

P937, Nasal cavity and Ethmoid sinuses, Lymphatic spread of nasal cavity primaries is uncommon.

Maxillary sinus, Suprastructure tumors extend into the nasal cavity,.... Invasion of these structures gives lesions of the suprastructure a poorer prognosis. Lymphatic spread of nasal cavity primaries is uncommon.

14. () 下列對 Laryngeal cancer 敘述，下列何者錯誤？

- A) 若腫瘤由 epiglottis 長出並侵犯到 base of the tongue 黏膜，病人有 vocal cord fixation, 依據 AJCC 第八版其原發腫瘤屬於 T4。

- B) Advanced vocal cord carcinoma 的首選治療是 total laryngectomy +/- adjuvant radiotherapy。
- C) Subglottic area 淋巴循環不豐富，而其淋巴會流向 level IV 和 VI 區域。
- D) RTOG 9111 study 發現，RT and concomitant cisplatin 對於 larynx preservation 會比 neoadjuvant chemotherapy followed by RT 好。

(14 答案): A (頭頸癌-Laryngeal cancer)

(14 難易度): 中等

(14 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 41, p1045-1050; 2019.

P1048, T4a Moderately advanced local disease; tumor invades through the thyroid cartilage and/or invades tissues beyond the larynx (e.g., trachea, soft tissues of the neck including deep extrinsic muscles of the tongue strap muscles, thyroid, or esophagus)

15. () 下列對 oral cavity cancer 敘述，下列何者錯誤？

- A) 根據 RTOG 9501 及 EORTC 22931 的綜合分析，extracapsular extension 及/或 microscopically involved surgical margin 是需要 CCRT 的最重要因素。
- B) Lip cancer regional metastasis 的機會較低，所以一般不需 elective nodal irradiation。
- C) Tumor size 和 depth of invasion 是決定舌癌是否有頸部淋巴結轉移的重要因子。
- D) Buccal cancer 通常制定原發腫瘤處 CTV 時範圍應較為廣泛，特別是 T3/T4 或 N2/N3 病灶時 masticator space and infratemporal fossa 要包含入 CTV。

(15 答案): B (頭頸癌-Oral cavity)

(15 難易度): 中等

(15 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, 7th edition; Chapter 41, p998-1001; 2019.

P1048, An important consideration in managing lip cancer is the risk of regional metastatic disease. Generally, the risk of regional lymph node metastatic disease for T1 and T2 cancers of the lip is lower than for stage-matched tumors of other oral cavity sites. Thus, END is recommended for patients with T3 and T4 carcinomas of the lip; however, it may not be warranted for all T1 and T2 lesions